

Assignment–14.2

Name:K.Tejaswini

Batch:21

Ht.no:2303A51425

Lab14:WebDesignApplication–AI-AssistedHTML/CSS/JS Generation

LabObjectives

- Designfunctional,visuallyappealingwebapplicationsusing HTML, CSS, and JavaScript with AI assistance.
- Applyresponsive,accessible,andinteractivedesign principles.

Week7- Tuesday

- CreatepracticalUIcomponentsforreal-worldweb applications.
- UseAltooptimize layout,UX,andperformance.

TaskDescription1–(LandingPage Design)

- Createasimplelandingpageforaproductorservice.Includea header, hero section, and footer.

Requirements:

- UseHTMLandCSStostructurethepage.
- UseAlto suggest layoutalignmentand typography.
- Ensurethepageisresponsiveacrossdevices.

PROMPT:“Generateasimple,responsivelandingpagewithheader,herosection, and footer using HTML & CSS. Use modern typography and clean layout.”

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Simple Landing Page</title>

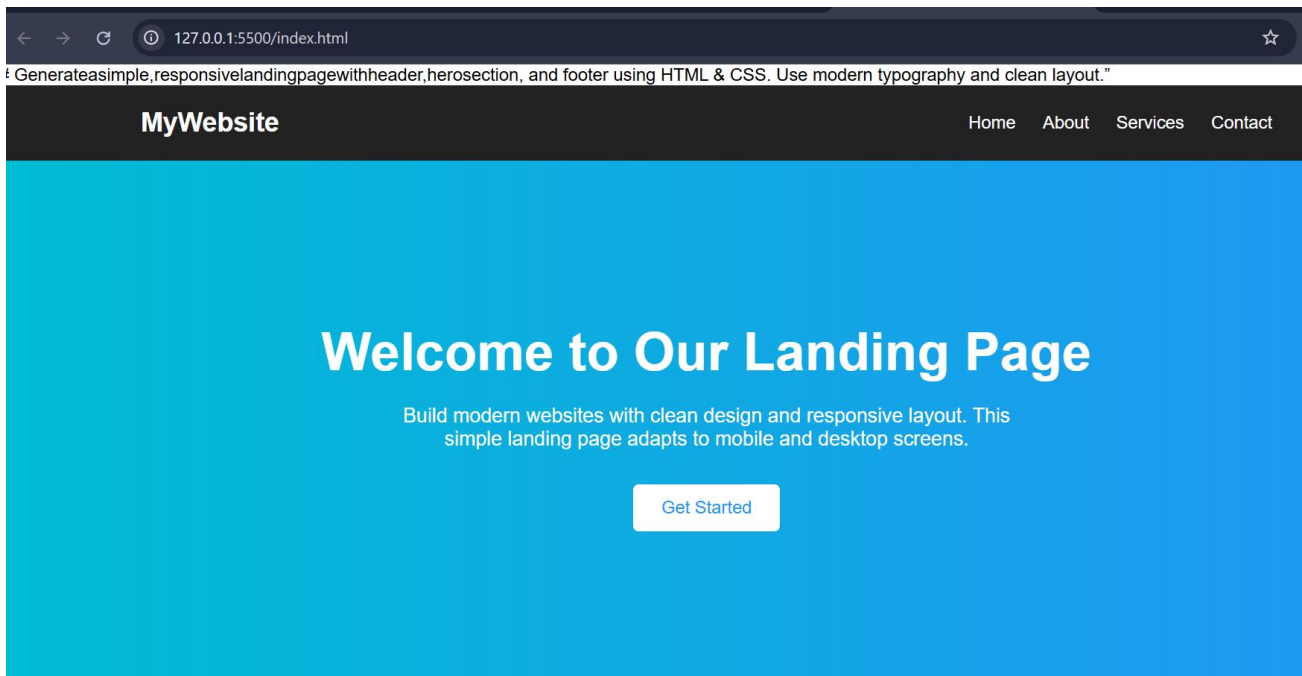
<style>

/* Reset */
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family: Arial, Helvetica, sans-serif;
}

/* Header */
header{
display:flex;
justify-content:space-between;
align-items:center;
padding:20px 10%;
background: #222;
color: white;
}

nav a{
color: white;
text-decoration:none;

```



1. Observation: The AI generated a clean and responsive web page layout using semantic HTML tags like header, section, and footer.
2. It automatically applied modern design patterns such as spacing, typography alignment, and color consistency.
3. AI used media queries to ensure responsiveness across mobile and desktop screens.
4. This task shows how AI simplifies UI structuring and reduces manual design effort.

Task Description-2 (Theme Toggle Feature)

- Provide a button to switch between light and dark themes..

Requirements:

- Use JavaScript to toggle CSS classes.
- Use AI assistance to manage color contrast and accessibility.
- Ensure theme preference applies instantly on click.

Prompt:

“Add a JavaScript button that toggles between light and dark theme by switching CSS classes.

```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Landing Page with Theme Toggle</title>
7
8  <style>
9
10 /* Default Light Theme */
11 body{
12 margin:0;
13 font-family:Arial, sans-serif;
14 background:■ white;
15 color:□ black;
16 }
17
18 /* Header */
19 header{
20 display:flex;
21 justify-content:space-between;
22 align-items:center;
23 padding:15px 10%;
24 background:□ #222;
25 color:■ white;
26 }
27
28 /* Hero Section */
29 .hero{
30 height:80vh;

```

##Add JavaScript theme toggle button that switches light/dark mode instantly by toggling CSS classes."

Toggle Theme

Welcome to the Theme Toggle Example

Click the button above to switch between light and dark mode.

Observation:

1. The AI implemented a simple and effective light/dark theme switch using

JavaScript class toggling.

-
2. It ensured proper color contrast between themes, improving accessibility and readability.
3. The code is minimal, efficient, and provides instant visual feedback to the user.

4. This demonstrates AI's ability to handle interactive UI behavior with clean logic.

Task Description 3—(User Feedback Form)

- Create a feedback form with fields: Username, Rating, Comments.

Requirements:

- Use JavaScript to validate required fields.
- Display success or error messages dynamically.
- Use AI to improve form usability and layout.

Prompt:

“Create a feedback form with validation and dynamic success/error messages using JavaScript.”

```
#Create a user feedback form with JavaScript validation. Fields: Username, Rating, Comments. Show success/error dynamically.
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Feedback Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 20px;
    }
    .form-group {
      margin-bottom: 15px;
    }
    .error {
      color: red;
    }
    .success {
      color: green;
    }
  </style>
</head>
<body>

  <h1>User Feedback Form</h1>
  <form id="feedbackForm">
    <div class="form-group">
      <label for="username">Username:</label>
      <input type="text" id="username" name="username">
    </div>
    <div class="form-group">
```

#"Create a user feedback form with JavaScript validation. Fields: Username, Rating, Comments. Show success/error dynamically."

User Feedback Form

Username:

Rating (1-5):

Comments:

Observation:

1. Alcreatedastructuredfeedbackformwithclearinputfieldsand placeholders for better usability.
2. ItaddedJavaScriptvalidationtopreventemptyorincorrectsubmissions.
3. Dynamicsuccess/errormessagesimproveuserinteractionandmakethe form more user-friendly.
4. ThistaskhighlightsAI'scapabilityingeneratingpractical,user-centered form handling logic.

TaskDescription-4(ImageGallerywith JavaScript)

- Task:Displayasmallimagegalleryonawebpage.

Requirements:

- UseJavaScripttodynamicallyloadimages.
- Addnext/previousnavigationbuttons.
- UseAltooptimize layoutresponsiveness.

Prompt: "Developanimagegallerythatswitchesimagesusingnextand previous buttons."

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Image Gallery</title>

<style>

body{
font-family: Arial, sans-serif;
text-align:center;
background: #f4f4f4;
margin:0;
padding:0;
}

h1{
margin-top:20px;
}

.gallery{
margin-top:40px;
}

img{
width:400px;
height:250px;
border-radius:10px;
box-shadow:0 4px 10px rgba(0,0,0,0.3);
```


Simple Image Gallery



Previous

Next

Observation:

1. The AI designed a functional image gallery using an array and next/previous navigation logic.
2. Modulo operations ensure smooth image cycling without errors.
3. Responsive image styling ensures the gallery works well on all screen sizes.
4. This shows AI's ability to implement dynamic UI components using clean and optimized JavaScript.

Task Description-5:(Notification Banner)

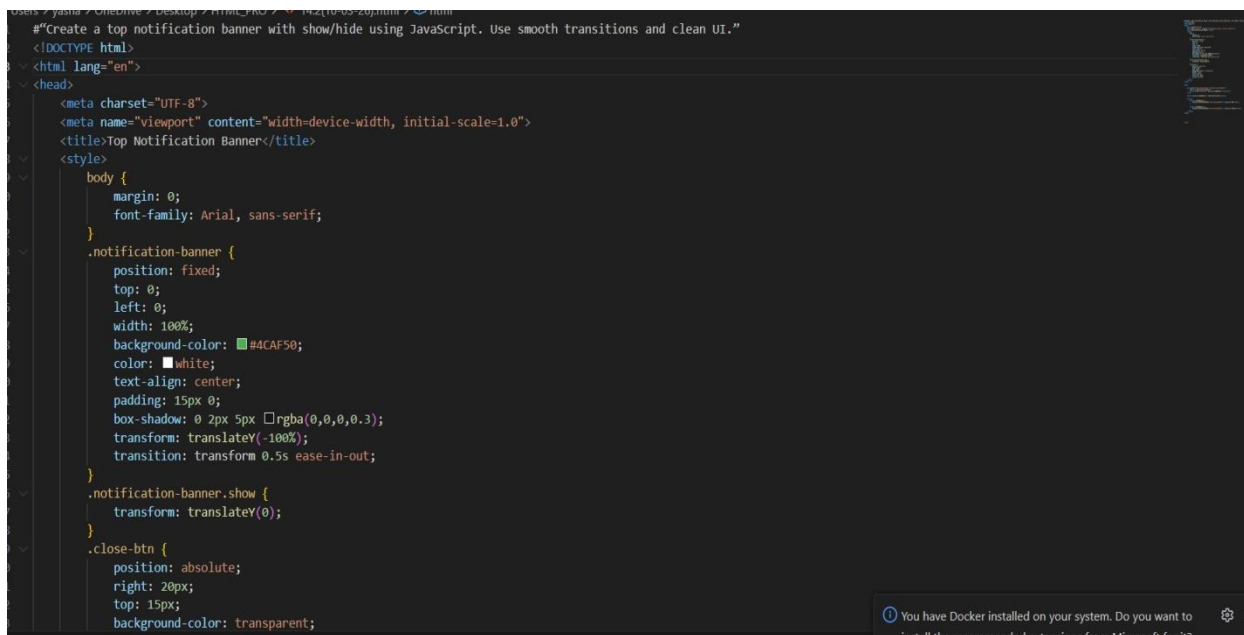
- Display a notification banner at the top of the page.

Requirements:

- Use JavaScript to show and hide the banner.
- Style the banner using CSS.
- Use AI to ensure smooth transitions and clean UI behavior.

Prompt:

“Create a notification banner with show/hide functionality and smooth animation.”



```
#"Create a top notification banner with show/hide using JavaScript. Use smooth transitions and clean UI."
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Top Notification Banner</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }
    .notification-banner {
      position: fixed;
      top: 0;
      left: 0;
      width: 100%;
      background-color: #4CAF50;
      color: white;
      text-align: center;
      padding: 15px 0;
      box-shadow: 0 2px 5px rgba(0,0,0,0.3);
      transform: translateY(-100%);
      transition: transform 0.5s ease-in-out;
    }
    .notification-banner.show {
      transform: translateY(0);
    }
    .close-btn {
      position: absolute;
      right: 20px;
      top: 15px;
      background-color: transparent;
```

This is a notification banner!



Observation:

1. AI created a visually appealing banner with a smooth hide animation using CSS transitions.
2. The mechanism uses class-based toggling for better code maintainability.
3. The banner displays the message clearly without affecting user interaction flow.
4. This task demonstrates AI's capability to integrate UI micro-interactions efficiently.

